

10(a)	<i>any eight from:</i> 1 reduced, NAD / FAD ; 2 passed to ETC ; 3 inner membrane / cristae ; 4 hydrogen released (from reduced, NAD / FAD) ; 5 split into electrons and protons ; 6 electrons pass along, ETC / carriers / cytochromes ; 7 energy released pumps protons into intermembrane space ; 8 proton gradient is set up ; 9 protons <u>diffuse</u>, (back) through membrane / down gradient ; 10 ATP synthase / stalked particles / protein channels ; 11 (ATP produced from) ADP and (inorganic) phosphate ; 12 <i>idea of</i> oxygen as final electron acceptor ; 13 addition of proton (to oxygen) to form water / (oxygen) reduced to water ;	8
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Question	Answer	Marks
10(b)	<i>any seven from:</i> 1 detail of placing larvae in respirometer ; e.g. on gauze / use spoon 2 use and role of, KOH / AW; 3 water bath / maintain temperature ; 4 choose 4 different temperatures ; 5 mark initial position of liquid drop ; 6 leave for set time ; 7 measure distance moved by drop (in set time) ; 8 (use tap to) reset respirometer ; 9 (at least 3) replicates ; 10 calculate mean values ; 11 calculate rate as distance ÷ time ; 12 plot graph of rate v temperature ;	7